

# Power BI: A Complete Introduction to Business Intelligence, Its Uses, Companies, and the Role of a Power BI Developer

## Introduction

In today's digital world, organizations generate enormous amounts of data every day. Sales transactions, customer information, financial records, employee data, operational metrics, website activity, market data, and many other types of information are continuously created and stored. However, **data itself does not automatically create business value**. Organizations need to transform that data into meaningful information and insights that people can use to make better decisions.

This is where **Microsoft Power BI** plays an important role.

Power BI is Microsoft's business intelligence and analytics platform. It enables organizations to connect to data from many different sources, transform and model that data, create interactive reports and dashboards, analyze trends, and share insights with business users. Microsoft describes Power BI as a platform that helps organizations turn data into actionable insights.

In simple terms:

**Power BI takes data and turns it into information that people can understand, analyze, and use to make decisions.**

For example, instead of giving a manager a large Excel file containing millions of sales transactions, a Power BI developer can create a dashboard showing total sales, profit, sales by region, top-performing products, monthly trends, and areas requiring attention.

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## What Is Power BI?

Power BI is a collection of Microsoft technologies designed for **business intelligence, data visualization, reporting, analytics, and decision-making**.

The Power BI ecosystem includes several important components.

### 1. Power BI Desktop

Power BI Desktop is primarily used by report developers and analysts to build reports and semantic models.

A developer can use Power BI Desktop to:

- Connect to databases
- Import Excel and CSV files
- Connect to cloud platforms

- Connect to APIs and web sources
- Clean and transform data
- Build relationships between tables
- Create a data model
- Write DAX calculations
- Design interactive reports
- Analyze data

Microsoft describes Power BI Desktop as the main environment for data modeling and report creation.

## 2. Power Query

Power Query is the data preparation and transformation engine used within Power BI.

For example, raw data may contain:

- Duplicate records
- Missing values
- Incorrect data types
- Unnecessary columns
- Different date formats
- Multiple files requiring consolidation

Power Query can clean and transform this data before it enters the analytical model.

Power Query uses the **M language** behind the scenes.

## 3. Data Modeling

Data modeling is one of the most important parts of professional Power BI development.

A developer creates relationships between tables and establishes a logical structure for analytical reporting.

For example:

### Sales Fact

- Product Dimension
- Customer Dimension
- Date Dimension
- Region Dimension

A well-designed model often follows a **star schema**, which improves usability, maintainability, and performance.

## 4. DAX

**DAX (Data Analysis Expressions)** is the formula language used in Power BI.

Developers use DAX to create measures and calculations such as:

- Total Sales
- Total Profit
- Profit Margin
- Year-to-Date Sales
- Previous-Year Sales
- Sales Growth %
- Customer Count
- Average Revenue

For example:

```
Total Sales = SUM(Sales[SalesAmount])
```

A simple formula can become extremely powerful when combined with Power BI's data model and filter context.

## 5. Power BI Service

After creating a report in Power BI Desktop, developers can publish it to the **Power BI Service**, Microsoft's cloud-based environment.

The Power BI Service supports:

- Workspaces
- Reports
- Dashboards
- Semantic models
- Apps
- Scheduled refresh
- Security
- Collaboration
- Alerts
- Subscriptions
- Deployment and administration

Microsoft explains that Desktop is primarily focused on report creation and modeling, while the Power BI Service supports sharing and collaboration.

## 6. Power BI Mobile

Power BI reports and dashboards can also be consumed through mobile applications, allowing users to monitor business information using phones and tablets.

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# Why Do Companies Use Power BI?

The fundamental reason companies use Power BI is **better decision-making**.

Imagine a company has information spread across:

- SQL Server
- Snowflake
- Excel
- SharePoint
- Salesforce
- APIs
- Azure
- Databricks
- CSV files

Without a BI platform, users may need to manually combine information and prepare reports.

Power BI can bring information from these sources together and present it through a common analytical layer.

This provides several major benefits.

### **1. Faster Decision-Making**

Instead of waiting for manually prepared reports, management can access dashboards and analyze information interactively.

### **2. Data Visualization**

Humans can understand patterns much more easily through appropriate visualizations than through millions of rows of raw data.

Power BI provides charts, tables, maps, KPIs, cards, decomposition trees, drill-through functionality, and many other visual capabilities.

### **3. Self-Service Analytics**

Power BI enables business users to explore data themselves rather than depending entirely on IT teams for every analytical question.

### **4. Centralized Reporting**

Organizations can create standardized reports and distribute them through Power BI workspaces and apps.

### **5. Security**

Enterprise organizations can control who can access particular data. Power BI supports capabilities such as **Row-Level Security (RLS)**, sensitivity labels, audit capabilities, and workspace governance.

## 6. Integration with Microsoft Ecosystem

Power BI integrates with Microsoft technologies such as:

- Microsoft Fabric
- Azure
- Excel
- Teams
- SharePoint
- Power Automate
- Power Apps

Microsoft now positions Power BI as a core workload within **Microsoft Fabric**.

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## Which Companies Use Power BI?

Power BI is used across many industries, including:

- Banking
- Financial services
- Insurance
- Retail
- Healthcare
- Manufacturing
- Telecommunications
- Technology
- Government
- Education
- Energy
- Logistics

Microsoft has published customer examples involving organizations such as **Walmart, T-Mobile, Bayer, H&M, Trek Bicycles, and RAC of Western Australia**.

For example, Walmart has used Power BI for large-scale financial reporting, including semantic models and advanced Power BI capabilities. T-Mobile has also been described by Microsoft as having hundreds of Power BI use cases.

Power BI is therefore not simply a tool for small Excel-based reports. It can be used as part of large enterprise BI environments.

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## Power BI and London Stock Exchange Group (LSEG)

An excellent example of Power BI's relevance to the financial industry is **London Stock Exchange Group (LSEG)**.

LSEG is a global financial markets infrastructure and data provider. Its business involves enormous volumes of financial and market data.

Microsoft and LSEG announced a **10-year strategic partnership** focused on transforming financial markets through cloud, data, analytics, AI, and Microsoft technologies. The partnership specifically identified technologies including **Azure, Synapse, Power BI, Excel, and Teams** as part of next-generation analytics and modeling solutions.

Power BI is also directly relevant to LSEG's data ecosystem. LSEG's developer community provides examples showing how **LSEG Data Platform data can be retrieved and visualized using Power BI**, including integration through Power Query and Python.

LSEG also publishes learning material around **Power BI dashboards for commodity-market analysis**, demonstrating practical applications of Power BI with financial and market data.

This demonstrates an important concept:

**Power BI can be used not only for traditional corporate reporting but also for analyzing complex financial and market information.**

Within an organization such as LSEG, Power BI can support areas such as operational reporting, financial analysis, workforce analytics, technology performance, risk reporting, project management, and other business intelligence requirements.

Recent LSEG-related roles also explicitly mention Power BI development, DAX, Power Query, data modeling, SQL, Snowflake, Fabric, dashboards, and reporting governance, illustrating the importance of Power BI skills within enterprise analytics environments.

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## Who Is a Power BI Developer?

A **Power BI Developer** is a professional who transforms business requirements and raw data into reliable analytical solutions using Power BI.

A Power BI Developer is not simply someone who creates charts.

A professional Power BI Developer typically works across several layers:

**Business Requirement**

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**Data Sources**

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**Data Engineering / Data Preparation**

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**Data Model**

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**DAX**

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**Power BI Report**

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**Power BI Service**

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**Security & Governance**

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**Business Decision**

A developer needs to understand both **technology and business requirements**.

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## What Does a Power BI Developer Do?

A Power BI Developer may perform many different responsibilities.

### Requirement Gathering

The developer talks with business stakeholders to understand questions such as:

- What information do you need?
- Who will use the report?
- What KPIs are important?
- How frequently should the data refresh?
- What level of detail is required?
- Who should be allowed to see the data?

### Data Connection

The developer connects Power BI to appropriate sources.

Examples include:

- SQL Server
- Oracle
- Snowflake
- Azure
- Databricks
- Excel
- SharePoint
- APIs
- CSV
- Microsoft Fabric

## **Data Transformation**

Power Query is used to clean and prepare the data.

## **Data Modeling**

The developer creates relationships and designs an efficient semantic model.

## **DAX Development**

The developer creates business calculations and measures.

## **Report Development**

The developer designs interactive dashboards and reports.

## **Performance Optimization**

Professional developers analyze report performance and optimize:

- Data models
- DAX
- Power Query
- Data sources
- Relationships
- Visuals
- Refresh processes

## **Security**

Developers may implement:

- Row-Level Security
- Workspace permissions
- App permissions
- Data access rules



## Deployment and Governance

A developer may also work with:

- Development environments
  - Test environments
  - Production environments
  - Deployment pipelines
  - Workspaces
  - Governance
  - Monitoring
  - Version control practices
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## Harsha Sri Sampath — An Example of a Power BI Professional

**Harsha Sri Sampath** can be viewed as an example of how the role of a modern Power BI professional extends beyond simply creating dashboards.

A professional working with Power BI in an enterprise environment such as LSEG can combine:

**Power BI + SQL + Data Modeling + DAX + Power Query + Python + Snowflake + Microsoft Fabric + Databricks + Business Analysis**

This combination is significantly more powerful than dashboard development alone.

For example, consider a business requirement:

"Management wants to understand why operational performance has declined."

A Power BI professional should not simply create a line chart.

Instead, the professional should:

1. Understand the business problem.
2. Identify the relevant data sources.
3. Extract the required data.
4. Clean and transform the data.
5. Build an appropriate semantic model.
6. Create DAX measures.
7. Analyze trends and patterns.
8. Design an intuitive dashboard.
9. Apply security.
10. Publish the report.
11. Configure refresh.
12. Monitor performance.
13. Explain the insights to stakeholders.

This is why modern Power BI professionals are increasingly becoming **analytics engineers, BI developers, data analysts, and business intelligence specialists**, rather than simply "report developers."

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## Power BI and Microsoft Fabric

Power BI has also become increasingly connected with **Microsoft Fabric**.

Microsoft Fabric is Microsoft's unified analytics platform, bringing together capabilities for:

- Data engineering
- Data integration
- Data science
- Real-time analytics
- Data warehousing
- Business intelligence

Power BI is one of the core workloads within Fabric.

A modern enterprise architecture could therefore look like:

### Data Sources

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### Data Engineering

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### Fabric Lakehouse / Warehouse

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### Semantic Model

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### Power BI

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### Reports & Dashboards

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### Business Users

This architecture allows organizations to move from raw data to business insight through a more integrated analytics platform.

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## The Future of Power BI

Power BI is also becoming increasingly connected with **AI and Copilot**.

Microsoft is integrating Copilot capabilities into the Fabric and Power BI ecosystem to help users explore data, explain insights, and accelerate report and semantic-model development.

This does not mean that Power BI developers will disappear.

Instead, the role is likely to evolve.

The future Power BI professional will need to understand:

**BI + Data Engineering + Analytics + AI + Business Domain Knowledge**

A developer who understands only visualization may become less competitive.

A developer who understands the complete data-to-insight lifecycle can provide much greater value.

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## Conclusion

Power BI is much more than a dashboarding tool. It is a comprehensive business intelligence and analytics platform that enables organizations to connect data, prepare it, model it, analyze it, visualize it, secure it, and distribute insights across an organization.

Companies use Power BI because they need to turn large amounts of complex data into understandable information and ultimately into better decisions.

Organizations such as Walmart and T-Mobile demonstrate large-scale Power BI adoption, while LSEG provides an especially relevant example for the financial-services industry, where Power BI can be combined with financial data, cloud technologies, analytics, and Microsoft's broader technology ecosystem.

The **Power BI Developer** sits at the center of this process. The role requires much more than creating attractive charts. A strong Power BI Developer understands business requirements, data sources, ETL, Power Query, data modeling, DAX, visualization, performance, security, deployment, and governance.

For professionals such as **Harsha Sri Sampath**, developing expertise across Power BI, SQL, Python, Snowflake, Databricks, and Microsoft Fabric creates a pathway from traditional BI development toward modern enterprise analytics and data engineering.

Ultimately, the purpose of Power BI can be summarized in one sentence:

**Power BI transforms data into insights, and insights into better business decisions.**